PhysAgent: Physics-Informed SDevice Code Generation via Ontology-Guided LLM

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***Abstract*—As semiconductor scaling reaches its limits, TCAD simulation becomes essential for DTCO. However, the core challenge lies in configuring SDevice physics models and parameters, not in structure generation. We propose PhysAgent, which builds a Sentaurus TCAD guide-based ontology and integrates literature-driven domain knowledge to generate physically grounded SDevice files without relying on synthetic training data.**

***Keywords*—TCAD, ontology, physics-informed modeling, parameter file generation, domain knowledge, semiconductor materials**

I. Introduction

Recent LLM-based TCAD approaches rely on supervised fine-tuning (SFT) with synthetic datasets, focusing primarily on SDE structure generation. However, publicly available TCAD data is scarce, and physics model selection remains implicit. Rather than fine-tuning domain-specific models, we argue that building a knowledge structure for general-purpose LLMs is more effective. Table I summarizes the key differences.

TABLE I
Comparison of SFT-based approaches and PhysAgent.

	SFT-based	**PhysAgent**
Training data required	Large synthetic	None
Target scope	SDE structure	SDevice physics
Physics model selection	Implicit / fixed	Explicit (ontology)
Material-specific knowledge	Not addressed	Literature-driven
LLM dependency	Domain-specific	General-purpose
Knowledge updatability	Retrain needed	Ontology update

We propose PhysAgent, which combines a Sentaurus TCAD guide-based ontology with literature-driven domain knowledge to generate physically grounded SDevice files for specific semiconductor materials.

II. Related Work

TCAD simulation consists of SDE for device geometry and SDevice for physics models and solver settings. While LLMs have been applied to SDE code generation through fine-tuning or in-context learning, SDevice configuration—selecting and parameterizing models such as mobility, recombination, and bandgap narrowing—remains unaddressed due to its high expertise requirements.

III. Proposed Framework

PhysAgent combines a TCAD physics ontology and literature-driven domain knowledge to guide LLM-based SDevice file generation (Fig. 1).

A. TCAD Physics Ontology

We construct a directed graph from the Sentaurus TCAD guide with three node types (*equation*, *parameter*, *constant*) and three edge types (*input*, *specialCase*, *pairwise*), capturing inter-model dependencies such as bandgap narrowing affecting intrinsic carrier density and recombination rates (Fig. 2).

B. Literature-Driven Domain Knowledge

The ontology captures model structure but not material-specific expertise. PhysAgent supplements it with literature-driven knowledge—validated model combinations and calibrated parameters for specific materials (e.g., SiC, GaN).

C. Ontology-Guided Generation

Given a target device and material, PhysAgent extracts a relevant ontology subgraph, combines it with literature knowledge, and provides this as structured context to the LLM, producing physically justified SDevice configurations.

D. Ontology Expansion and Validation

Unlike SFT-based approaches requiring retraining for new models or materials, PhysAgent uses a directional synchronization workflow (Fig. 3) with three artifact layers—Spec, Plan, and Impl—where changes propagate automatically from the source of truth while reverse modifications require explicit approval, ensuring incremental expansion without retraining.

IV. Result

V. Conclusion

We presented PhysAgent, which generates physically grounded SDevice files by combining a TCAD guide-based ontology with literature-driven domain knowledge, enabling general-purpose LLMs to support expert-level TCAD modeling without synthetic training data.

References

[1] SISPAD Comittee, "SISPAD Website". [Online]. Available: https://sispad2026.jp/

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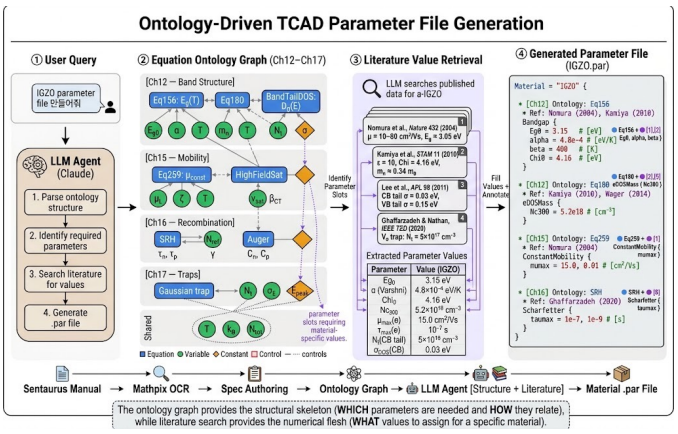

Fig. 1. PhysAgent framework overview.

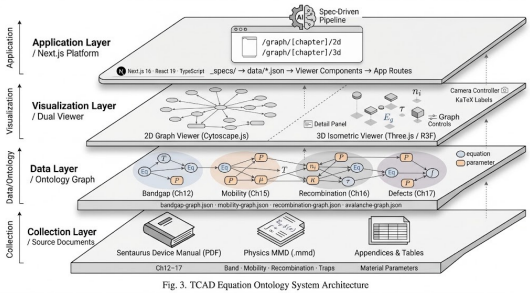

Fig. 2. Ontology system architecture.

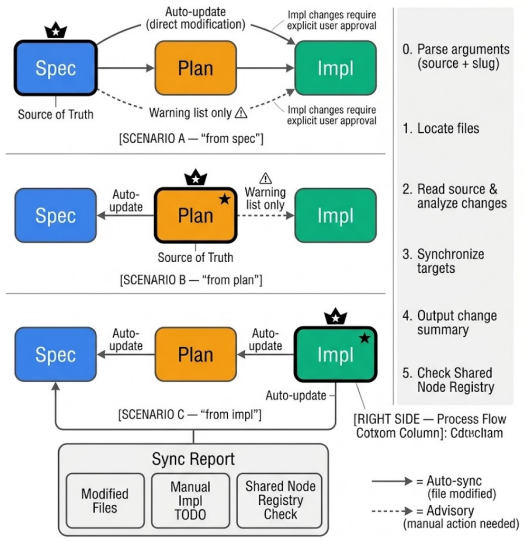

Fig. X. Directional synchronization workflow. The user designates one artifact as the source of truth; dependent artifacts are updated automatically (solid arrows) or flagged for manual review (dashed arrows). Implementation files are never auto-modified unless explicitly requested, preserving the safety-first principle.

Fig. 3. Expansion and validation workflow.